Unveiling Cybercrime Patterns in India: A Clustering-Based Geographic and Visual Analytical Approach

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***Abstract*— Cybercrime is a growing concern that affects both individuals and institutions in India due to the country's rapid digitization of services and increased internet penetration. This study uses unsupervised learning techniques to present a data-driven analysis of cybercrime trends in Indian States and Union Territories from 2017 to 2020. Two clustering algorithms, K-Means and Spectral Clustering are used to find patterns in cybercrime risk, detect evolving hotspots and identify the dominant cybercrime motives across different regions. Clustering performance evaluation shows that Spectral Clustering performs better overall, with an accuracy of 0.92, precision of 0.86, recall of 0.99 and f1-score of 0.92, making it more efficient in fully identifying high-risk areas. Conversely, K-Means shows a marginally higher precision of 1.00, indicating that it finds fewer but more precise hotspots. These insights are intended to help cybersecurity experts, law enforcement organizations, and policymakers create more targeted, area-specific interventions, better allocation of resources, and improve digital safety infrastructure.**

Keywords—cybercrime, K- Means clustering, Spectral clustering, crime hotspots, risk assessment

I. Introduction

Cybercrime has been on the rise in India for a very long time now, because of its ambitious digital transformation initiatives, such as Digital India, expanding internet penetration to over 760 million users by 2020, and widespread smartphone adoption. While there are many positive effects of the digital revolution on the economy and society, there are vulnerabilities and opportunities for cybercriminals to take advantage of. The attack surface for cybercriminals has increased due to the growth of digital payment systems, e-commerce platforms, social media networks, and online banking services. The range of cybercrimes has expanded greatly, encompassing everything from intricate financial fraud schemes to identity theft, online harassment, and data breaches. Conventional crime patterns have changed into intricate digital representations that call for new analytical techniques to comprehend their underlying patterns, temporal trends, and geographic distribution.

II. Literature Review

Through the identification of trends, clusters, and anomalies over time and space, crime pattern analysis is an essential component of comprehending and reducing crime activities.Before the era of machine learning, numerous studies have investigated crime analysis through spatial and visual methods. Craglia et al. [1] used GIS-based techniques to examine crime trends in Sheffield in order to show how spatial tools can be applied to law enforcement. CrimAnalyzer, a visual analytics tool that allows for temporal tracking of crime dynamics in Sao Paulo was introduced by Garcia et al. [2]. Brantingham et al. suggested a fine-grained, bottom-up method that used individual land parcels and fuzzy topology to uncover more complex crime patterns using the data from British Columbia [3].

Later, with the emergence of machine learning techniques, in a number of studies, clustering techniques for crime pattern analysis have been investigated. The classification of crimes as heinous, non-heinous and special/local law violations using K-Means on Delhi Police FIR data was successfully done in [4] by Gera and Vohra. The research concentrated on the prevailing types of crime in Delhi and their spread over different area categories such as slums, residential areas and commercial hubs. K-Means clustering was used by Kigerl [5] to analyze cybercrime data from 190 countries using metrics such as malware, internet usage, etc., this research categorized nations into four typologies — low cybercrime, non-serious cybercrime, advance-fee fraud, and phishing-dominant countries.

A hotspot detection technique utilizing the K-Means and Expectation-Maximization algorithms was proposed in [6] by Vineet et al., where the work emphasized geospatial plotting of crime and the pragmatic applicability of clustering to assist law enforcement personnel by enhancing response time and investigative efficiency. Sivaranjini et al used DBSCAN, agglomerative Clustering and K-Means to compare the crime trends of six major Tamil Nadu cities over a 15-year period. The objective was to find the best clustering model for predicting crime, where K-Means was mapped using Google Maps for better interpretability [7]. P. Y. Lee et al. [8] used Spectral Clustering and Mean Shift to find spatial patterns in more than 450,000 records of Los Angeles crime data from 2010-2019. The study focused on hotspot identification and spatial patterns of crime for optimal law enforcement efficiency and resource allocation. In addition to clustering based techniques, recent studies have used advanced machine learning and computer vision techniques for cybercrime and anomaly detection. To examine cybercrime patterns, Sridharan et al. [9] and Kumaran et al. [10] used predictive modeling and data mining. Conversely, Ramya et al. [11] and Kanishkar et al. [12] concentrated on real-time surveillance through neural network architectures which enhance proactive crime monitoring systems.

Prior research works have used clustering techniques to examine physical crimes in particular areas but they mostly ignore cybercrime on a national scale. For instance, prior research did not concentrate on cybercrime in Indian States and Union Territories (UTs), instead it concentrated on crime profiling, hotspot mapping, or urban crime trends using techniques like DBSCAN and Mean Shift. The present study,

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on the other hand, focuses on pan-India cybercrime analysis using Spectral and K-Means Clustering.

III. Problem Statement

Despite improved awareness about the emerging cyber threats, a better understanding of the patterns, trends and regional variations of cybercrime across various States and UTs is still lacking [13],[14]. One of the key challenges with regards to this is the need to categorize these regions in terms of their level of cybercrime risk. This will help in prioritizing resources and interventions accordingly, as cybercrime is an emerging risk. The lack of identification of cybercrime hotspots in India also reduces the capability of the authorities to take targeted measures against cybercrime thus hampering its successful prevention.

The paper aims to bridge these gaps by leveraging clustering techniques to uncover hidden patterns and reveal regional variations in cybercrime behavior. The present research attempts to provide in-depth analysis of the States and UTs cybercrime patterns for the period 2017-2020 in India using unsupervised learning methods such as K-Means and Spectral Clustering.

IV. Proposed Methodology

In order to accurately analyze cybercrime patterns across Indian States/UT and establish significant patterns, step-by-step methodology is used. The flowchart in Fig. 1 illustrates the significant steps used — from data collection to clustering and mapping risk areas:

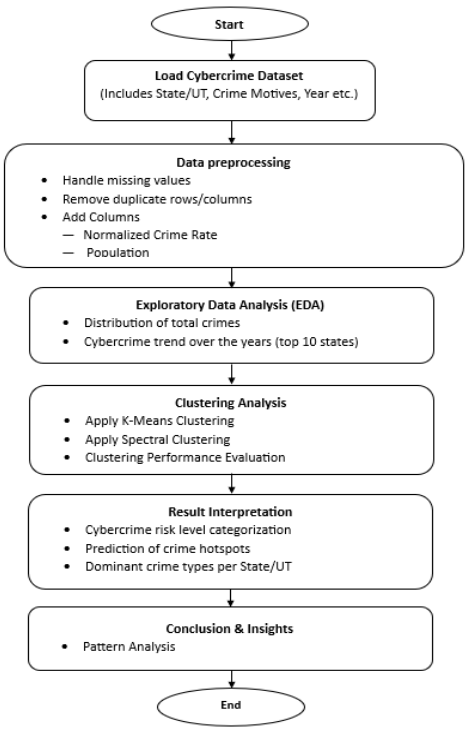

Fig. 1. Methodology flowchart implemented for crime pattern analysis

A. Data Collection and Preprocessing

The study makes use of a cybercrime dataset that is obtained from the openly accessible Kaggle platform. A comprehensive picture of cybercrime trends from 2017 to 2020 is provided by the dataset which spans 36 States/UTs' and has 23 columns covering different cybercrime motives.

Raw data is pre-processed multiple times to prepare it for clustering. Non-relevant columns are dropped to remove redundancy. New features are introduced to make the dataset more strong analytically, the Population column is generated from official census data, Normalized Crime Rate (NCR) is calculated as the total cybercrimes per population (100,000 people).

$$NCR = \frac{Total\ Cybercrimes}{Population} \times 100{,}000 \quad (1)$$

All the categorical variables are encoded properly, and numerical columns are scaled using Standard Scaler to generate balanced input for clustering algorithms.

B. Exploratory Data Analysis

Exploratory data analysis is an important step where the primary features of a dataset are summarized and visualized. The analysis process involved looking at how all reported crimes are distributed and observing the trends of cybercrime among the top 10 contributing States/UTs.

C. Clustering Analysis

In this study, two types of primary clustering methods are used: K-Means and Spectral Clustering.

Unsupervised techniques are chosen over deep learning because the data is unlabeled, the emphasis is on finding regional patterns, and computationally efficient models are required. Without requiring complicated training, these techniques are ideal for evaluating tabular data and identifying significant groupings.

D. Result Interpretation

Using K-Means and Spectral Clustering, the analysis produced three main findings:

1.) All the States/UTs are categorized into risk categories (Low, Moderate, High and Very High) .
2.) Geospatial visualisation highlight locations with the highest concentrations of cybercrime (hotspots).
3.) To guide policy actions, dominant cybercrime motives are identified.

These outcomes provide a comprehensive picture of the intensity, spread, and nature of cybercrime across India.

V. Analysis of Cybercrime Patterns

K-Means and Spectral Clustering is performed to serve three fundamental analytical goals:

Case 1: Categorization of Risk Level for Cybercrime (2017-2020)

- Features Considered – Latitude, Longitude, Normalized Crime Rate.
- No. of Clusters – The value of K is considered as 4. States/UTs are grouped into four risk categories ranging from low to very high
- Risk Level Categorization
  - Low Risk: Cluster with the lowest average crime rate.
  - Moderate Risk: Cluster with an average crime rate in the middle.
  - High Risk: Cluster with a significantly high crime rate.
  - Very High Risk: Cluster with the highest average crime rate.

Case 2: Identification of Crime Hotspots (2017-2020)
The aim is to identify cybercrime hotspot areas in India for every year from 2017 to 2020.

- Features Considered – Latitude, Longitude, Total number of cybercrime cases per State/UT for each year.
- No. of Clusters – The value of K is considered as 2 to distinguish high-threat areas (hotspot) and low-threat areas (non-hotspot)
- Hotspot Identification
  - Hotspot: Cluster with a higher mean crime count.
  - Non-Hotspot: Cluster with a lower mean crime count.

Case 3: Prediction of dominant Cybercrime motives by State/UT (2017-2020)
The aim is to classify and group States/UTs on the basis of common patterns of cybercrime motives and predict the most dominant form of cybercrime for each State/UT for the years 2017-2020.
Features Considered – There are 18 types of cybercrime motives captured in the dataset which are grouped under 8 categories :

  - Personal motives: Personal Revenge, Anger, Sexual Exploitation.
  - Financial motives: Fraud, Extortion.
  - Social motives: Causing Disrepute, Prank, Abetment to Suicide
  - Political and ideological motives: Political Motives, Terrorist Activities, Inciting Hate Against Country.
  - Economic motives: Developing Own Business, Sale/Purchase of Illegal Drugs
  - Technical motives: Steal Information, Spreading Piracy, Disrupt Public Service.
  - Psychological motives: Psycho or Pervert
  - Miscellaneous: Others
- No. of Clusters – The value of K is considered as 3, based on the elbow method and these clusters represents states grouped on the basis of similar crime motive patterns.
- Prevalent Crime type prediction – For every State/UT, the most prominent type of crime is found by checking the motive of crime which has the greatest number of cases.

VI. Results and Discussion

The section provides the most important findings that have been obtained from exploratory data analysis and clustering methods used on the cybercrime dataset.

A. Exploratory Insights of Cybercrime Data

Using the combination of various visual techniques like bar plots, and line charts, this research reveals the dynamic scenario of trends of cybercrime in India from 2017 to 2020.

In India, the number of cybercrime cases more than doubled from about 2,000 in 2017 to over 50,000 in 2020 as represented in Fig. 2.

According to the trend analysis in Fig. 3, Uttar Pradesh, Karnataka and Maharashtra reported higher number of crimes. Assam experienced rapid growth after 2019 ,whereas states such as Rajasthan, Jharkhand, and Odisha reported lower number of crimes. Between 2018 and 2019, the majority of States/UTs saw notable increase.

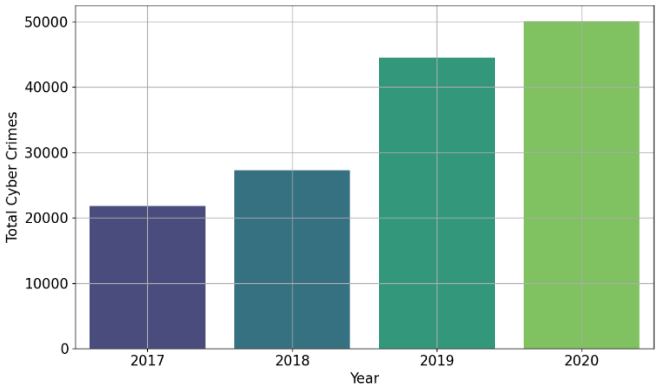

Fig. 2. Total Cyber Crime cases in India per year (2017 – 2020)

The insights of the exploratory analysis based on visualizations are a primary point to understand how cybercrime is changing in India.

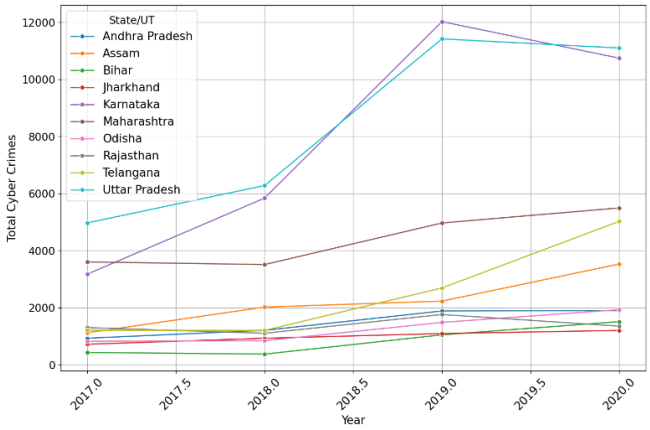

Fig. 3. Trend of Cyber Crimes for top 10 states (2017-2020).

B. Crime Pattern Analysis using K-Means Clustering

K-Means clustering is performed for the following three cases:

For each case, visualizations for all the years (2017-2020) produced comparable patterns and insights, the output from 2020 is only shown as a representative example in Fig. 4 to Fig. 6.

Case-1: Categorization of Risk Level for Cybercrime from 2017 to 2020
K-Means segmented the States/UTs into four risk levels based on the crime intensity, with each risk graphically indicated by a different color for easy distinction:

- Low Risk- Green
- Moderate Risk- Orange
- High Risk- Red
- Very High Risk- Dark Red

The results of clustering are visualized using Folium which is an interactive mapping library allowing for intuitive geospatial analysis. The representation of cybercrime risk levels for 2020 is shown in Fig.4.

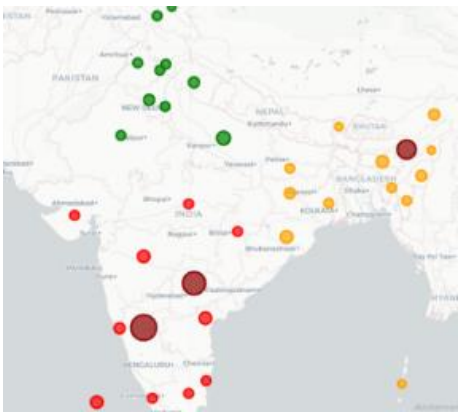

Fig. 4. States/UTs' grouped by risk level for 2020 (K-Means)

The key patterns observed in this visualisation of risk levels are :

1.) Persistant High-Risk Areas: Telangana and Karnataka stayed in the 'Very High Risk' category, indicating high prevalance of cybercrime.
2.) States with Rising Risk Trends: Kerala and Andhra Pradesh moved from 'Moderate'/'Low' risk in 2017 to 'High Risk' by 2020.
3.) Stable Low-Risk Zone: North-Eastern States like Assam, Manipur, Meghalaya, Mizoram and Nagaland were consistently classified as 'Low' or 'Moderate Risk' areas, indicating either underreporting or limited exposure.

Case 2: Identification of Crime Hotspots from 2017 to 2020

K-Means is applied to classify the regions in India into hotspots and non-hotspots for each year (2017-2020) mainly based on mean crime count. Visual identification of hotspot for 2020 is done as in Fig.5.

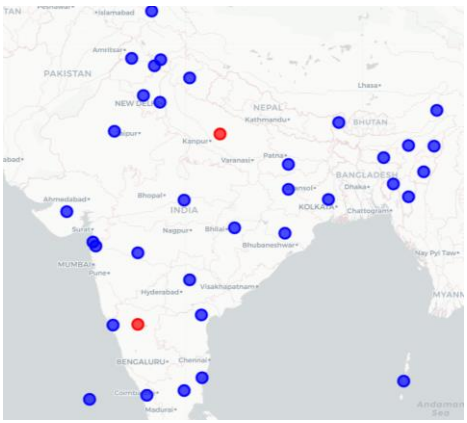

Fig. 5. Identification of hotspots for 2020 (K-Means)

- Hotspot- Red
- Non-Hotspot- Blue

Karnataka and Uttar Pradesh are identified as a cybercrime hotspot over all the years (2017-2020). Maharashtra is identified as a hotspot only in 2017 but not in the later years.

Case 3: Prediction of dominant cybercrime motive from 2017 to 2020

Most Prevalent cybercrime motives across the States/UTs' from 2017 to 2020 are determined. The representation of year 2020 is shown in Fig.6.

Fig. 6. Prediction of dominant cybercrime motives for 2020

TABLE I lists the color codes which are used to represent different cybercrime motives in the visualization for 2020.From the data presented, the following are some of the key inferences that can be drawn regarding the dominant cybercrime trends in States/UTs':

- Leading cybercrime: Fraud is the common motive in 26 out of 36 States/UTs' by 2020, particularly in major cities like Delhi, Maharashtra, Karnataka and Gujarat.
- Notable changes in crime trends over time: In Tamil Nadu and Kerala, the focus shifted from fraud to other crimes like disrepute and sexual exploitation.
- Temporal trends: By 2020, sexual exploitation was more common than crimes like piracy and political motivation.

TABLE I COLOR MAPPING FOR CYBERCRIME MOTIVES

Motive	Color
Fraud	Orange
Sexual Exploitation	Pink
Terrorist Activities	Dark Blue
Disrupt Public Service	Cyan
Others	Light Gray

In majority of States/UTs, fraud continues to be the most common cybercrime, especially in economically vibrant areas. Yet, the appearance of other motives like sexual exploitation and disrepute causing in certain areas for some years indicates changing cybercrime patterns and vulnerabilities in society, thus emphasizing targeted interventions and ongoing monitoring.

C. Crime Pattern Analysis using Spectral Clustering

Spectral clustering is also applied across the same three cases as in K-Means clustering:For each case, visualizations for all the years (2017-2020) produced comparable patterns and insights, the output from 2020 is only shown as a representative example similar to K-Means clustering as shown in Fig. 7 and Fig. 8.

Case-1: Categorization of Risk Level for Cybercrime from 2017 to 2020

The color coding and the mapping technique are same as in K-Means clustering. It is observed that Spectral Clustering is able to identify different groupings by making use of graph-based relationships. The representation of cybercrime risk levels using Spectral clustering for 2020 is shown in Fig.7.

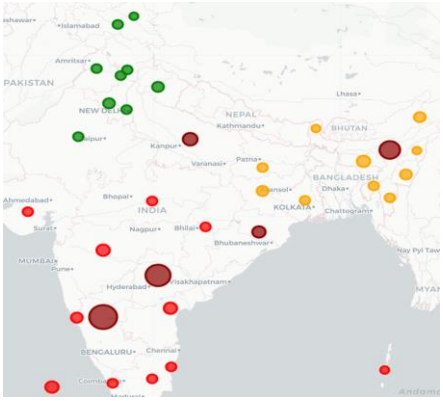

Fig. 7. States/UTs' grouped by risk level for 2020 (Spectral)

The key patterns observed in the visualisation of risk levels are:

1.) Consistent high risk states: The maps point out Andhra Pradesh, Karnataka, Maharashtra, Uttar Pradesh and Telengana appeared under 'Very High Risk' level throughout the whole duration.

2.) Fluctuating risk levels: These maps underscore the dynamic changes in classification of risk levels possibly due to changes in policy, improvement in reporting mechanisms or actual changes in cybercrime activity.

- By 2020, Goa, Kerala and Puducherry showed a transition from 'Low'/'Moderate' to 'High'/'Very High Risk', indicating an increase in the severity of cybercrime.
- States like Delhi, Haryana, Punjab, Rajasthan and Chandigarh transitioned toward 'Moderate' or 'Low Risk' level by 2020.

Case 2: Identification of Crime Hotspots from 2017 to 2020

Besides K-Means, Spectral clustering is also applied to partition the regions in India into hotspots and non-hotspots for each year (2017-2020) mainly based on mean crime count.

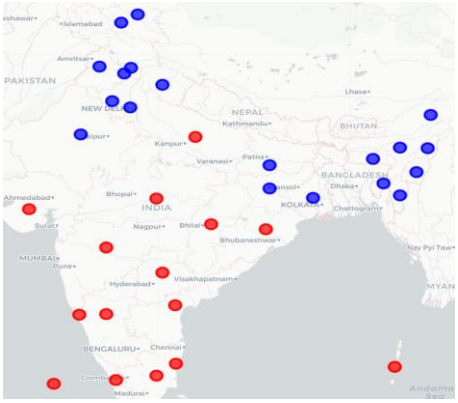

Fig. 8. Identification of hotspots for 2020 (Spectral)

15 hotspots are identified in 2017,2018 and 2020 while 14 hotspots are identified in 2019. The hotspots identified in 2020 is presented in Fig.8. The color code is same as used in K-Means method of clustering.

TABLE II shows that Spectral clustering can detect more crime hotspots than K-Means, as it detects more subtle similarities between areas by taking advantage of the underlying graph structure of the data

TABLE II COMPARISON OF HOTSPOTS DETECTED BY SPECTRAL AND K-MEANS CLUSTERING

Type of Hotspot	Spectral Clustering	K-Means Clustering
Consistent Hotspots (2017-2020): 12 States appeared as hotspots consistently	Andhra Pradesh, Karnataka, Uttar Pradesh, Telengana, Tamil Nadu, Chattisgarh, Kerala, Madhya Pradesh, Maharashtra, Odisha, Gujarat, Goa.	Karnataka, Uttar Pradesh
Intermittent Hotspots: States/UTs which appeared for some years	A & N Islands, Puducherry (2017,2018, 2020); Lakshadweep (2018-2020); D & N Haveli (2017)	Maharashtra (2017)

Case 3: Prediction of dominant cybercrime motive from 2017 to 2020

In contrast to the previous cases (Case 1 & Case 2), Spectral clustering produces same results as that of K-Means in this case.

Below are some of the possible reasons:

- Clear Feature Boundaries: The prevalent cybercrime motives exhibit clear-cut distinctions, thus, making it possible for both the algorithms to cluster efficiently.
- Linear Seperability: The data for the type of crimes is likely to be more linearly seperable than the above cases.
- Discrete Categories: Motives serve as unique labels which makes clustering more straightforward or easier for both the algorithms.

D. K-Means vs. Spectral Clustering: Performance Comparison

To evaluate the effectiveness of K-Means and Spectral Clustering in detecting cybercrime hotspots in Indian States /UTs, both the algorithms are being compared using conventional clustering evaluating measures:

- Accuracy- The ratio of accurately classified hotspot states and non-hotspot states to all states.
- Precision- The ratio of accurately predicted hotspot states to all the states which are predicted as hotspots.
- Recall- The ratio of true hotspot states correctly classified by the model.
- F1- Score- The harmonic mean of precision and recall.

TABLE III. PERFORMANCE COMPARISON OF K-MEANS AND SPECTRAL CLUSTERING

Clustering Method	Accuracy	Precision	Recall	F1-Score
K-Means	0.60	1.00	0.20	0.33
Spectral	0.92	0.86	0.99	0.92

The measures in TABLE III offer a complete perspective on each algorithm's capacity for accurately distinguishing between high risk (hotspot) and low risk (non-hotspot) areas.

Spectral clustering is observed to outperform K-Means in terms of accuracy, recall, and f1-score, thereby representing a balanced overall detection of hotspots. K-Means is observed to have a slightly higher precision than Spectral, indicating fewer but more accurate hotspot predictions. If thorough or comprehensive detection is required, then Spectral clustering is preferable. If high confidence in the detection of hotspots is required, then K-Means can be used.

VII. Conclusion

The paper explores the implementation of , both K-Means and Spectral clustering algorithms are used to examine the trends and geographic distribution of cybercrimes across Indian States and UTs'. The study successfully classified regions based on cybercrime risk levels, identified recurrent hotspots and revealed dominant cybercrime motives from 2017 to 2020 on normalized cybercrime data and geospatial attributes. The interpretation of clustering results is improved by the use of Folium-based visualizations, which successfully highlighted annual trends and regional variances. In view of the goal of this research to reveal larger regional cybercrime trends and support necessary early actions, Spectral Clustering proves to be the better and more credible method.

The study illustrates unsupervised machine learning methods and geospatial analysis can be used to understand India's growing cybercrime threat. Despite certain limitations such as data inconsistencies and algorithmic limitations, the project lays the groundwork for upcoming developments in real-time monitoring, predictive modeling and evidence-based policy. Furthermore, because the study ends in 2020, it could not include the post-COVID spike in digital activity and online dependence, which probably changed the dynamics of cybercrime. Post 2020 data can be used in future research to analyze shifts and create more adaptable and fact-based prevention and policy measures.

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